In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE (Main)-2017

CONCEPT RECAPITULATION TEST - II

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

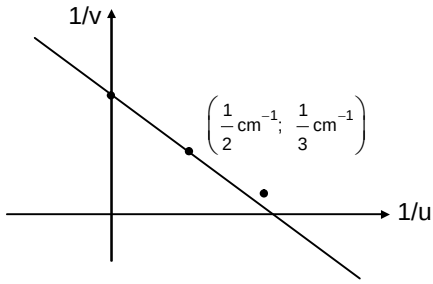
(One Option Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

1. The biding energy per nucleon for deuteron $\left({}_1^2H\right)$ and helium $\left({}_2^4He\right)$ are 1.1 MeV and 7.0 MeV respectively. The energy released when two deuterons fuse to form a helium nucleus is
   (A) 23.6 MeV (B) 47.12 MeV
   (C) 34.4 MeV (D) 11.8 MeV

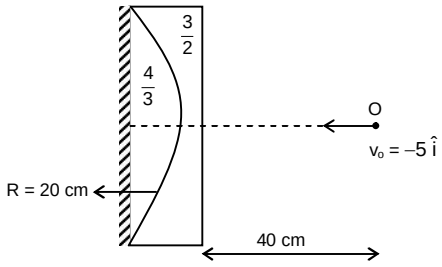
2. The K_α X-ray emission line of tungsten occurs at λ = 0.021 nm. The energy difference between K and L levels in this atom is about
   (A) 0.51 MeV (B) 1.2 MeV
   (C) 59 keV (D) 13.6 eV

3. $\frac{1}{v}$ vs $\frac{1}{u}$ graph for a spherical mirror is shown in the figure. The focal length of the mirror will be
   (A) 5 cm
   (B) –5 cm
   (C) $\frac{6}{5}$ cm
   (D) $-\frac{6}{5}$ cm



Space for Rough work

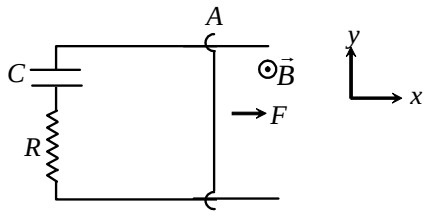
4. The lens combination having refractive indices $\frac{4}{3}$ & $\frac{3}{2}$ as shown in the figure, is silvered. The velocity of image will be?



(A) $1.8\,\hat{i}$ (B) $3\,\hat{i}$

(C) $3.6\,\hat{i}$ (D) $5\,\hat{i}$

5. A conducting rod *AB* moves parallel to *x*-axis in the *x*-*y* plane. A uniform magnetic field *B* pointing normally out of the plane exists throughout the region. A force *F* acts perpendicular to the rod, so that the rod moves with uniform velocity *v*. The force *F* is given by (neglect resistance of all the wires)



(A) $\frac{vB^2l^2}{R}e^{-t/RC}$ (B) $\frac{vB^2l^2}{R}$

(C) $\frac{vB^2l^2}{R}\left(1-e^{-t/RC}\right)$ (D) $\frac{vB^2l^2}{R}\left(1-e^{-2t/RC}\right)$

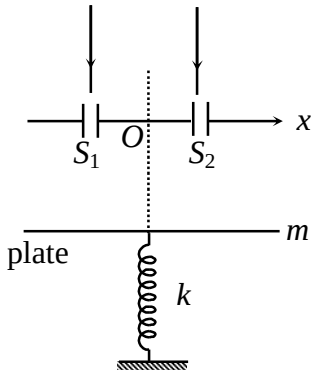
6. In a vernier calipers 9 divisions of it's main scale match with 10 divisions on its vernier scale. Each division of the main scale is of 20 units. Least count of apparatus is:

(A) 1 unit (B) 2 units

(C) 3 units (D) 4 units

Space for Rough work

7. Two slits S_1 and S_2 lie on the x-axis and symmetric with respect to y-axis are illuminated by a parallel monochromatic light beam of wavelength λ as shown. The distance between slits is d (>> λ). Point O is the mid point of the line S_1S_2 and this point is considered as the origin. The slits are in horizontal plane. The interference pattern is observed on a horizontal plate (acting as screen) of mass m which is connected to one end of a vertical massless spring of spring constant k. The other end of the spring fixed to ground. At t = 0, the plate is at a distance D (>> d) below the plane of slits and spring is in its natural length. The plate is released from rest from its initial position. The rate by which fringe, width will increase when acceleration of plate is zero, is



(A) $\frac{\lambda g}{d}\sqrt{\frac{m}{k}}$ (B) $\frac{\lambda g}{3d}\sqrt{\frac{m}{k}}$

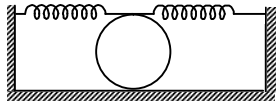
(C) $\frac{\lambda g}{4d}\sqrt{\frac{m}{k}}$ (D) $\frac{\lambda g}{2d}\sqrt{\frac{m}{k}}$

8. A perfectly absorbing surface intercepts a parallel beam of monochromatic light of power 100W (λ = 500 nm) incident on it normally. If the force exerted by light beam on the surface is $\frac{10^{-6}}{n}$ N, then n =

(A) 1 (B) 2

(C) 3 (D) 4

9. A solid sphere of mass *M* is attached to two massless springs each of spring constant *k.* It can roll without slipping along a horizontal surface. If the system is released after a small stretch in spring, then the time period of oscillations will be :



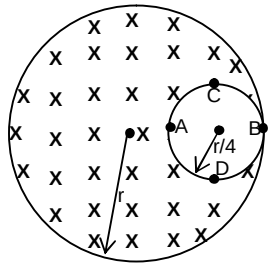
(A) $2\pi\sqrt{\frac{7M}{2k}}$ (B) $2\pi\sqrt{\frac{7M}{5k}}$

(C) $\pi\sqrt{\frac{7M}{10k}}$ (D) $2\pi\sqrt{\frac{M}{k}}$

Space for Rough work

10. A closed organ pipe of length *L* is vibrating in its first overtone. There is a point *Q* inside the pipe at a distance 7*L*/9 from the open end. The ratio of pressure amplitude at *Q* to the maximum pressure amplitude in the pipe is
(A) 1 : 2 (B) 2 : 1
(C) 1 : 1 (D) 2 : 3

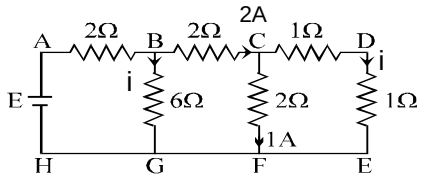
11. Consider a spherical charge distribution of volume charge density ρ c /m^3 Potential difference between A & B



(A) $\frac{2\rho}{3\varepsilon_0} r^2$ (B) 0

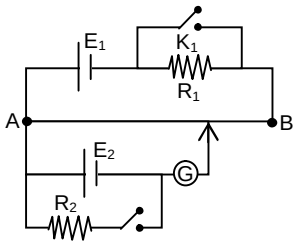
(C) $\frac{\rho}{4\varepsilon_0} r^2$ (D) $\frac{\rho r^2}{8\varepsilon_0}$

12. The figure shows a network of resistors and a battery. If 1A current flows through the branch CF, then the emf E of the battery is



(A) 24 V (B) 12 V
(C) 18V (D) 6V

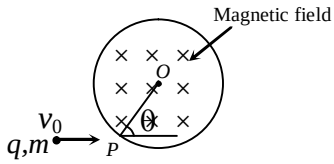
13. Figure shows the circuit of a potentiometer. The length of the potentiometer wire AB is 50 cm. The e.m.f. of the battery is 4 volt, having negligible internal resistance. Values of resistances R_1 and R_2 are 15 ohm and 5 ohm respectively. When both the keys are open, the null point is obtained at distance of 31.25 cm from end A but when both the keys are closed, the balance length reduces to 5 cm only. Given R_{AB} = 10 Ω. The e.m.f. of the cell E_2 is



(A) 1 volt (B) 2 volt
(C) 3 volt (D) 4 volt

Space for Rough work

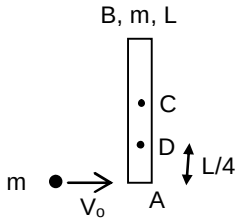
14. A particle charge *q* and mass m is projected with a velocity v_0 towards a circular region having uniform magnetic field *B* perpendicular and into the plane of paper from point *P* as shown in the figure. *R* is the radius and *O* is the centre of the circular region. If the line OP makes an angle θ with the direction of v_0 then the value of v_0 so that particle passes through *O* is



(A) $\frac{qBR}{m\sin\theta}$ (B) $\frac{qBR}{2m\sin\theta}$

(C) $\frac{2qBR}{m\sin\theta}$ (D) $\frac{3qBR}{2m\sin\theta}$

15. A thin, uniform rod of mass M and length L is at rest on a smooth horizontal surface. A particle of same mass M collides with the rod at one end perpendicular to its length with velocity V_o and sticks to it. C is the middle point of the rod and D is a point at a distance of L/4 from end A. The time taken by the rod to turn through 90° after collision is



(A) $\frac{L}{V_o}$ (B) $\frac{\pi L}{V_o}$

(C) $\frac{3\pi L}{5V_o}$ (D) $\frac{5\pi L}{12V_o}$

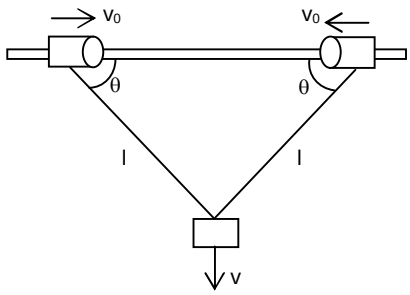
16. Velocity time equation of a particle moving in a straight line is $V = t^2 - 5t + 6$. The distance travelled by the particle in the time interval from t = 0 to t = 4 sec

(A) 0 (B) $\frac{17}{3}$

(C) 6 (D) $\frac{16}{3}$

Space for Rough work

17. In the shown figure two identical beads slide along a smooth horizontal rod as shown in figure. The relation between v and v_0 in the shown position will be



(A) $v = v_0 \cot\theta$

(B) $v = v_0 \sin\theta$

(C) $v = v_0 \tan\theta$

(D) $v = v_0 \cos\theta$

18. Water is flowing streamline motion through a horizontal tube. The pressure at a point in the tube is p where the velocity of flow is v. At another point, where the pressure is p/2, the velocity of flow is (density of water $= \rho$) :

(A) $\sqrt{v^2 + \frac{p}{\rho}}$ (B) $\sqrt{v^2 - \frac{p}{\rho}}$

(C) $\sqrt{v^2 + \frac{2p}{\rho}}$ (D) $\sqrt{v^2 - \frac{2p}{\rho}}$

19. A rod collides elastically with smooth horizontal surface after falling from a height. For maximum angular speed of the rod just after impact, the rod should be released in such a way that it makes an angle α with horizontal, the value of α will be

(A) 0° (B) $\cos^{-1}\frac{1}{\sqrt{2}}$

(C) $\cos^{-1}\frac{1}{\sqrt{3}}$ (D) $\cos^{-1}\frac{1}{\sqrt{6}}$

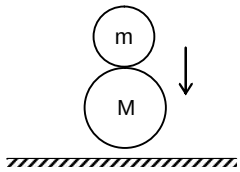
Space for Rough work

20. Three particles A, B and C are situated at the vertices of an equilateral triangle ABC of side d at t = 0, each of the particles moves with constant speed u. A always has its velocity along AB, B along BC and C along CA. Angular velocity of the line AB, at the time t, $\left(0 < t < \frac{2d}{3u}\right)$

(A) $\frac{u}{2d - 3ut}$ (B) $\frac{u\sqrt{3}}{2d - 3ut}$

(C) $\frac{2u}{d\sqrt{3} - 3\sqrt{3}\,ut}$ (D) none of these

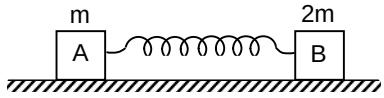
21. A small ball of mass m is placed on top of the "super ball" of mass M and the two ball are dropped to the floor from height h. Height rise of small ball after the collision is Nh, then 'N' is: (Assume that the collision with the superball are elastic, and the m << M)



(A) 8 (B) 9

(C) 10 (D) 11

22. In the figure shown, the spring is compressed by x_0 = 4 cm and released. Two locks A and B of masses m and 2 m respectively, are attached at the ends of the spring of force constant 4 N/m. Blocks are kept on a frictionless horizontal surface and released. The work done by the spring (in mJ) on A by the time compression of the spring is reduced to $\frac{x_o}{4}$ is:

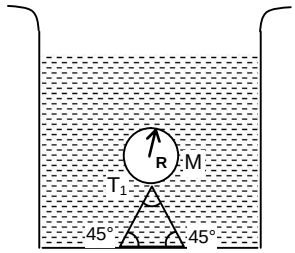


(A) 1 (B) 1.5

(C) 2 (D) 3

Space for Rough work

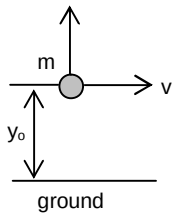
23. A hollow sphere of mass M and radius r is immersed in a tank of water (density ρ_w). The sphere would float if it were set free. The sphere is tied to the bottom of tank by two wires which makes angle 45° with the horizontal as shown in figure. The tension T_1 in the wire is



(A) $\frac{\frac{4}{3}\pi R^3\rho_w g - Mg}{\sqrt{2}}$ (B) $\frac{2}{3}\pi R^3\rho_w g - Mg$

(C) $\frac{\frac{4}{3}\pi R^3\rho_w g + Mg}{\sqrt{2}}$ (D) $\frac{4}{3}\pi R^3\rho_w g + Mg$

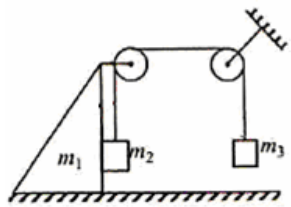
24. A particle of mass m is moving with a velocity of $v = \sqrt{\frac{k}{m}}$ in the horizontal direction at a height y_0 from the ground (y_0 = k/2mg). A variable force F = k/y acts on the particle at t = 0 in the vertical direction where y is the height of the particle from the ground. Find the speed of the particle when its acceleration become zero for the very first time. (k is a constant (N-m)



(A) $2\sqrt{\frac{k\,\ell n3}{m}}$ (B) $\sqrt{\frac{2k\,\ell n2}{m}}$

(C) $\sqrt{\frac{2k\,\ell n3}{m}}$ (D) $2\sqrt{\frac{k\,\ell n2}{m}}$

25. In the following figure both the pulleys and the string are massless and all the surfaces are frictionless.
Given: m_1 = 1 kg, m_2 = 2 kg, m_3 = 3 kg. The acceleration of m_1 is



(A) $\frac{40}{7}\,m/s^2$ (B) $\frac{30}{7}\,m/s^2$

(C) $\frac{20}{7}\,m/s^2$ (D) None of these

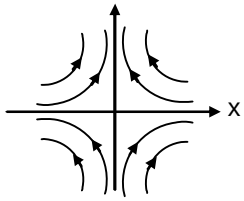
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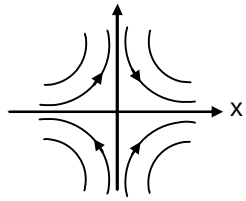
26. Potential energy of a particle moving along x-axis is given by $U = \frac{x^3}{3} - \frac{9x^2}{2} + 20x$. Find out position of stable equilibrium state.
(A) 3 (B) 5
(C) 6 (D) 7

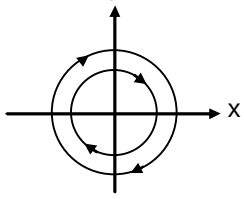
27. A conductor in the form of a right angle ABC with AB = 3 cm and BC = 4 cm carries a current of 10 A. There is a uniform magnetic field of 5 T perpendicular to the plane of the conductor. The force on the conductor will be
(A) 1.5 N (B) 2.0 N
(C) 2.5 N (D) 3.5 N

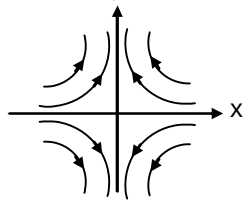
28. Curie temperature is the temperature above which:
(A) a ferromagnetic material becomes paramagnetic.
(B) a paramagnetic material becomes diamagnetic.
(C) a ferromagnetic material becomes diamagnetic.
(D) a paramagnetic material becomes ferromagnetic.

29. Potential is varying with x and y as $V = 2\left(x^2 - y^2\right)$. The corresponding field pattern is:

(A)


(B)


(C)


(D)


30. A tunnel is dug along diameter inside the surface of earth and a particle is projected from the centre of tunnel. The minimum velocity of particle such that it escape out from the earth's gravitation field is (Radius of earth = R_e)

(A) $\sqrt{2gR_e}$ (B) $\sqrt{\frac{3}{2}gR_e}$

(C) $\sqrt{3gR_e}$ (D) $\sqrt{\frac{5}{2}gR_e}$

Space for Rough work

Chemistry

PART – II

SECTION – A

(One Option Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

1. $KO_2 + S \xrightarrow{\Delta} (P) \xrightarrow{BaCl_2} (Q)\downarrow (\text{White ppt.})$

$(S)\uparrow \xleftarrow[\text{gas}]{HCl} (R) \xleftarrow[\Delta]{\text{Carbon}}$ (Q)

The unknown compounds P, Q, R and S in the above reaction sequence are respectively

(A) K_2SO_3, $BaSO_3$, BaS, H_2S (B) K_2S, BaS, CS_2, H_2S

(C) K_2SO_4, $BaSO_4$, BaS, H_2S (D) K_2SO_4, $BaSO_4$, $BaCO_3$, CO_2

2. $\boxed{\text{Borax}} \xrightarrow{HCl} (A) + NaCl$

$(C) \xleftarrow{\Delta} (B) \xleftarrow{H_2O}$ (A)

$\xrightarrow{Mg} \text{Boron} + (D)$

$\xrightarrow{HNO_3} (E) \xrightarrow{\Delta} (D) + (F)\uparrow + O_2$

Which of the above unknown compound forms an alkaline solution when reacts with water?

(A) C (B) D

(C) E (D) B

Space for Rough work

3. Which of the following compound is most acidic in nature?

(A) $CH_3\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2COOH$ (B) $CH_3CH_2\underset{\displaystyle CH_3}{\underset{|}{C}}HCOOH$

(C) $CH_3CH_2CH_2CH_2COOH$ (D) $H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-COOH$

4. Which of the following reaction produces nascent hydrogen?

(A) Passing of steam on red hot coke (B) Passing of steam on red hot iron
(C) Hydrolysis of ionic hydrides (D) Reaction of sodium with ethanol

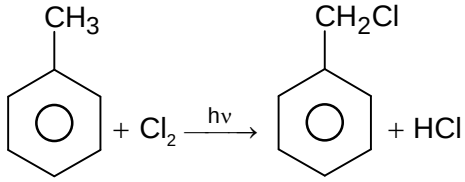
5. The smog is essentially caused by the presence of

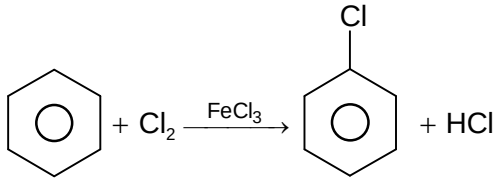
(A) O_2 and O_3 (B) O_2 and N_2
(C) SO_2 and NO_2 (D) O_3 and N_2

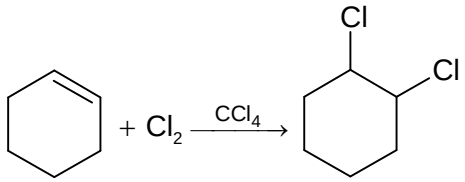
6. Which of the following carbonate needs maximum temperature for decomposition?

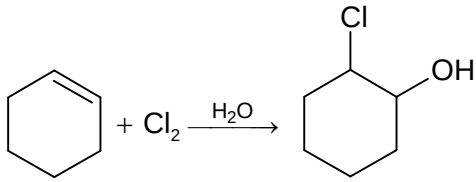
(A) Li_2CO_3 (B) Na_2CO_3
(C) $MgCO_3$ (D) $CaCO_3$

7. Which of the following is an electrophilic substitution reaction?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

8. The process of 'eutrophication' is due to
(A) increase in concentration of insecticide in water
(B) increase in concentration of fluoride ion in water
(C) the reduction in concentration of the dissolved oxygen in water due to phosphate in water
(D) attack of younger leaves of a plant by peroxyacetyl nitrate

9. Which of the following is the most abundant hydrocarbon pollutant?
(A) CH_4 (B) C_2H_6
(C) C_3H_8 (D) C_4H_{10}

10. In aqueous solution beryllium sulphate exists as:
(A) $[Be(H_2O)_4]SO_4$ (B) $[Be(H_2O)_6]SO_4$
(C) $[Be(H_2O)_3]SO_4$ (D) $[Be(H_2O)_5]SO_4$

11. Which of the following liquid has the highest boiling point?
(A) H_2O (B) H_2O_2
(C) HCl (D) NH_3

12. Which of the following ion does not cause hardness of water?
(A) Ca^{2+} (B) Mg^{2+}
(C) NO_3^- (D) HCO_3^-

13. The normal boiling point of a liquid 'X' is 500 K. Which of the following statement(s) is/are incorrect for the following process?
$X(\text{liquid}) \longrightarrow X(\text{gas})$
(A) At 500 K and 1 atm pressure, $\Delta G = 0$
(B) At 500 K and 1.5 atm pressure, ΔG = +ve
(C) At 500 K and 0.2 atm pressure, ΔG = -ve
(D) At 520 K and 1 atm pressure, ΔG = +ve

Space for Rough work

14. Which of the following complex(es) does not show(s) geometrical isomerism?
(A) $[Cr(H_2O)_5Cl]Cl_2.H_2O$ (B) $[Pt(NH_2CH_2COO)_2]$
(C) $[PtCl_2P(C_2H_5)_3]_2$ (D) $K_2[Fe(CN)_4(NH_3)_2]$

15. Find ΔH at 358 K for the reaction,
$Fe_2O_3(s) + 3H_2(g) \longrightarrow 2Fe(s) + 3H_2O(\ell)$
Given that, $\Delta H_{298} = -33.29$ kJ/mole and C_p for Fe_2O_3(s), Fe(s), $H_2O(\ell)$ and $H_2(g)$ are 103.8, 25.1, 75.3 and 28.8 J/K mole.
(A) 85.9 kJ/mole (B) –28.136 kJ/mole
(C) +28.136 kJ/mole (D) 85.9 J/mole

16. Heat of neutralization of four monobasic acids A, B, C and D are -13.7, - 9.4, - 11.2 and – 12.4 Kcal / mole respectively, when they are neutralized by a common base. The acidic character obeys the order
(A) A > B > C > D (B) A > D > C > B
(C) D > C > B > A (D) D > B > C > A

17. The amount of NaOH required to be dissolved in one litre of solution to give a pH of 11 at 25°C
(A)0.04 g/l (B) 0.4 g/l
(C) 0.01 g/l (D) 0.53 g/l

18. The number of α and β particles emitted in nuclear reaction $_{90}Th^{228} \longrightarrow {}_{83}Bi^{212}$ are respectively
(A) 4,1 (B) 3,7
(C) 8,1 (D) 4,7

19. A weak acid HX has the dissociation constant 1×10^{-5} M. It forms a salt NaX on reaction with alkali. The percentage hydrolysis of 0.1 M solution of NaX is
(A) 0.0001% (B) 0.01%
(C) 0.1% (D) 0.15%

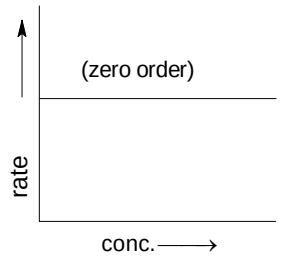
Space for Rough work

20. What is the molar solubility of $Mn(OH)_2$(Ksp = 4.5×10^{-14}) in a buffer solution containing equal amounts of NH_4^+ & NH_3(K_b=1.8×10^{-5})
(A) 3.0×10^{-5} (B) 1.38×10^{-4}
(C) 1.38×10^{-2} (D) 7.3×10^{-6}

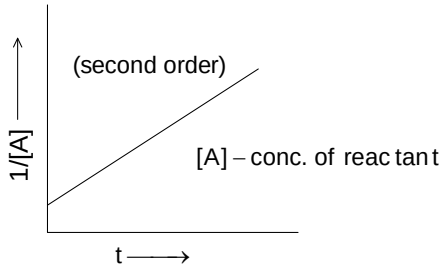
21. For the reaction,
$C_7H_8(l) + 9O_2(g) \rightarrow 7CO_2(g) + 4H_2O(l)$, calculated heat of reaction is 232.6 kJ/mol and observed heat of reaction is 50.4 kJ/mol, then the resonance energy is
(A) – 182.2 kJ / mol (B) + 182.2 kJ / mol
(C) 172 kJ/ mol (D) None of these

22. Select the incorrect graph

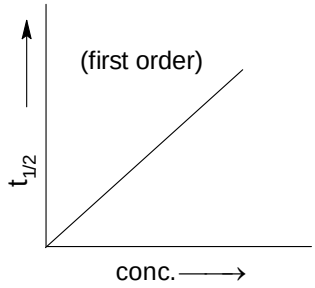
(A)



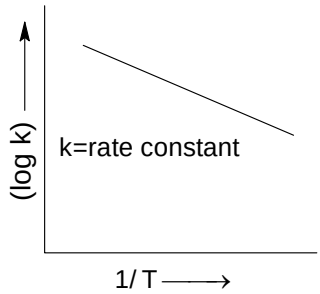
(B)



(C)



(D)



Space for Rough work

23. The following two equilibria exist simultaneously in a closed container of volume 2 L

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

$A_2(g) + C(g) \rightleftharpoons A_2C(g)$

Which of the following change does not affect the concentration of AB(g)

(A) Increase in pressure (B) Addition of C

(C) Addition of He gas at constant volume (D) All of these

24. An acid-base indicator has $K_a = 3 \times 10^{-5}$. The acid form of the indicator is red and the basic form of the indicator is blue. By how much must the pH change in order to change the indicator from 75% red to 75% blue? [log 3 = 0.4770]

(A) 4.05 (B) 0.95

(C) 5 (D) 9

25. For an elementary reaction

$aA + bB \longrightarrow P$; $\frac{dx}{dt} = k[A]^a.[B]^b$. If concentration of A is doubled, rate is doubled. If B is doubled, rate becomes four times. Which is correct?

(A) $\frac{-d[A]}{dt} = \frac{-d[B]}{dt}$ (B) $\frac{-d[A]}{dt} = \frac{-2d[B]}{dt}$

(C) $\frac{-2d[A]}{dt} = \frac{-d[B]}{dt}$ (D) none of these

26. In borax bead test, which compound is formed?

(A) Meta borate (B) Ortho borate

(C) Tetra borate (D) Double oxide

Space for Rough work

27. $A + C_2H_5OH \xrightarrow{H_2SO_4} (B)$, B on ignition gives green colour. A could be

(A) salt of sodium
(B) salt of iron
(C) salt of boric acid
(D) salt of meta phosphoric acid

28. AlF_3 is most soluble in

(A) HF
(B) KF
(C) KHF_2
(D) F_2

29. Solubility of $PbCl_2$ in 0.5 M NaCl is S_1 and that in 0.5 M $Pb(NO_3)_2$ is S_2 then which of the following is correct

(A) $\frac{S_1}{S_2^2} = 1$
(B) $S_1 - S_2^2 > 0$
(C) $S_1 - S_2^2 < 0$
(D) $\frac{S_1}{S_2} < 0$

30. Which silicate is formed from $[SiO_4]^{4-}$ tetrahedral units by sharing three oxygen atoms?

(A) Sheet silicate
(B) Pyro silicate
(C) Linear chain silicate
(D) 3-Dimensional silicate

Space for Rough work

Mathematics

PART – III

SECTION – A

(One Option Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

1. Circles of radii 5, 5 and 8 are mutually externally tangent. If a fourth circle of radius r touches all the three circles externally, then the value of r is

(A) 1 (B) $\frac{7}{8}$

(C) $\frac{8}{9}$ (D) $\frac{9}{11}$

2. If α,β are the roots the quadratic equation $x^2-\left(3+2^{\sqrt{\log_2 3}}-3^{\sqrt{\log_3 2}}\right)x-2\left(3^{\log_3 2}-2^{\log_2 3}\right)=0$, then the value of $\alpha^2+\alpha\beta+\beta^2$ is equal to

(A) 11 (B) 7

(C) 3 (D) 5

3. If $f(x)=\int_0^x\left(e^t+3\right)dt$, then which of the following is always true $\forall\, x\in R$?

(A) $f(x-3)-f(x-1)\le 11$ (B) $f(x+1)-f(x-1)<4$

(C) $f(x+3)-f(x-5)>24$ (D) $f(x)+f(x+1)<-21$

4. If e_1 and e_2 are the roots of the equation $x^2-ax+2=0$, where e_1 and e_2 are the eccentricities of an ellipse and hyperbola, respectively, then the value of a belongs to

(A) $(3,\infty)$ (B) $(2,\infty)$

(C) $(1,\infty)$ (D) $(-\infty,1)\cup(1,2)$

Space for Rough work

5. If $\theta_1 = \sin^{-1}\frac{4}{5} + \sin^{-1}\frac{1}{3}$ and $\theta_2 = \cos^{-1}\frac{4}{5} + \cos^{-1}\frac{1}{3}$, then

(A) $\theta_1 < \theta_2$ (B) $\theta_1 > \theta_2$

(C) $\theta_1 = \theta_2$ (D) None of these

6. $lx + my = 1$ is the equation of the chord PQ of $y^2 = 4x$ whose focus is S. If PS and QS meet the parabola again at R and T, respectively, then the slope of RT is

(A) $-\frac{1}{m}$ (B) $\frac{1}{m}$

(C) $\frac{2}{m}$ (D) None of these

7. Let ABC be an equilateral triangle, let KLMN be a rectangle with K, L on BC, M on AC and N on AB. Suppose AN/NB=2 and the area of triangle BKN is 6. The area of the triangle ABC is

(A) 54 (B) 108

(C) 48 (D) not determinable with the above data

8. The number of solutions in isosceles triangle ABC, $\tan A + \tan B + \tan C = \lambda, \lambda < 0$, is

(A) 0 (B) infinite

(C) 1 (D) 2

9. $\int_0^{\pi/2} \sin x \sin 2x \sin 3x \sin 4x \, dx =$

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{8}$

(C) $\frac{\pi}{16}$ (D) $\frac{\pi}{32}$

Space for Rough work

10. If $\sqrt{x+y}+\sqrt{y-x}=c$ then $\frac{d^2y}{dx^2}$ equals

(A) $\frac{2}{c^2}$ (B) $\frac{-2}{c^2}$

(C) $\frac{2}{c}$ (D) $\frac{-2}{c}$

11. If the curves $x^2-y^2=4$ and $xy=\sqrt{5}$ intersect at points A and B, then the possible number of point(s) C on the curve $x^2-y^2=4$ such that triangle ABC is equilateral is

(A) 0 (B) 1

(C) 2 (D) 4

12. If $A+B+C=\pi$ and $\sin\left(A+\frac{C}{2}\right)=k\sin\frac{C}{2}$, then $\tan\frac{A}{2}\tan\frac{B}{2}$ is equal to

(A) $\frac{k-1}{k+1}$ (B) $\frac{k+1}{k-1}$

(C) $\frac{k}{k+1}$ (D) 1

13. The number of rectangles that can be obtained by joining four of the twelve vertices of a 12 – sided regular polygon is

(A) 66 (B) 30

(C) 24 (D) 15

14. If $z^2+(p+iq)z+r+is=0$ where p, q, r, s are non – zero, has real roots, then

(A) $pqs=s^2+q^2r$ (B) $pqr=r^2+p^2s$

(C) $prs=q^2+r^2p$ (D) $prs=p^2+s^2q$

Space for Rough work

15. Let $a > 1$ be a real number and $f(x) = \log_a x^2$ for $x > 0$. If f^{-1} is the inverse function of f and b and c are real numbers then $f^{-1}(b+c)$ is equal to

(A) $f^{-1}(b).f^{-1}(c)$ (B) $f^{-1}(b) + f^{-1}(c)$

(C) $\dfrac{1}{f(b+c)}$ (D) $\dfrac{1}{f^{-1}(b) + f^{-1}(c)}$

16. The number of points at which $g(x) = \dfrac{1}{1+\dfrac{2}{f(x)}}$ is not differentiable, where $f(x) = \dfrac{1}{1+\dfrac{1}{x}}$ is

(A) 1 (B) 2
(C) 3 (D) 4

17. Cosine of the acute angle between the curves $y = 3^{x-1}\ln x$ and $y = x^x - 1$ is

(A) 0 (B) 1

(C) $\dfrac{\sqrt{3}}{2}$ (D) $\dfrac{1}{2}$

18. If $A_1, A_3, ...A_{2n-1}$ are n skew symmetric matrices of same order then

$B = \sum_{r=1}^{n} (2r-1)(A_{2r-1})^{2r-1}$ will be

(A) symmetric (B) skew symmetric
(C) neither symmetric nor skew symmetric (D) data is inadequate

19. A class has three teacher. Mr. X. Ms. Y and Mrs. Z and six students A, B, C, D, E, F. Number of ways in which they can be seated in a line of 9 chairs, if between any two teachers there are exactly two students, is

(A) 18.6! (B) 12.6!
(C) 24.6! (D) 6.6!

Space for Rough work

20. If $x=\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+.....y=\frac{1}{1^2}+\frac{3}{2^2}+\frac{1}{3^2}+\frac{3}{4^2}.....$ and $z=\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+.....$, then

(A) x, y, z are in A.P.

(B) $\frac{y}{6},\frac{x}{3},\frac{z}{1}$ are in A.P.

(C) $\frac{y}{6},\frac{x}{3},\frac{z}{2}$ are in A.P.

(D) $6y,3x,2z$ are in H.P.

21. In the xy – plane, the length of the shortest path from (0, 0) to (12, 16) that does not go inside the circle $(x-6)^2+(y-8)^2=25$ is

(A) $10\sqrt{3}$

(B) $10\sqrt{5}$

(C) $10\sqrt{3}+\frac{5\pi}{3}$

(D) $10+5\pi$

22. Suppose $y=f(x)$ and $y=g(x)$ are two functions whose graphs intersect at the three points (0, 4), (2, 2) and (4, 0) with $f(x)>g(x)$ for $0<x<2$ and $f(x)<g(x)$ for $2<x<4$. If $\int_0^4[f(x)-g(x)]dx=10$ and $\int_2^4[g(x)-f(x)]dx=5$, the area between two curves for $0<x<2$, is

(A) 5

(B) 10

(C) 15

(D) 20

23. If A is square matrix of order 2×2 such that $A^2=I; B=\begin{bmatrix}1 & \sqrt{2}\\ 0 & 1\end{bmatrix}$ and $C=ABA$, then $(C)^{2009}=$

(A) $A\begin{bmatrix}1 & 2009\sqrt{2}\\ 0 & 1\end{bmatrix}$

(B) $A\begin{bmatrix}1 & 2009+\sqrt{2}\\ 0 & 1\end{bmatrix}A$

(C) $A\begin{bmatrix}1 & \sqrt{2}\\ 0 & 2009\end{bmatrix}A$

(D) $A\begin{bmatrix}1 & 2009\sqrt{2}\\ 0 & 1\end{bmatrix}A$

Space for Rough work

24. $\lim\limits_{x\to\infty} x^2 \sin\left(\log_e \sqrt{\cos\frac{\pi}{x}}\right) =$

(A) 0 (B) $-\frac{\pi^2}{2}$

(C) $-\frac{\pi^2}{4}$ (D) $\frac{\pi^2}{8}$

25. Let $I_1 = \int_0^\infty \frac{x^2\sqrt{x}}{(1+x)^6}dx, I_2 = \int_0^\infty \frac{x\sqrt{x}}{(1+x)^6}dx,$ then

(A) $I_1 = 2I_2$ (B) $I_2 = 2I_1$

(C) $I_1 = I_2^2$ (D) $I_1 = I_2$

26. Let AB be the chord of contact of the point (5, –5) w.r.t. the circle $x^2 + y^2 = 5$. Then the locus of the orthocenter of the $\Delta PAB,$ where P be any point moving on the circle, is

(A) $(x-1)^2 + (y+1)^2 = 5$ (B) $(x-1)^2 + (y+1)^2 = \frac{5}{2}$

(C) $(x+1)^2 + (y-1)^2 = 5$ (D) $(x+1)^2 + (y-1)^2 = \frac{5}{2}$

27. In the interval $\left[0, \frac{\pi}{2}\right]$, the equation $\cos^2 x - \cos x - x = 0$ has

(A) no solution (B) exactly one solution

(C) exactly two solutions (D) more than two solutions

Space for Rough work

28. If $\tan\left(\sum_{r=1}^{n}\tan^{-1}\left(\frac{2r-1}{\left(r^2+r+1\right)\left(r^2-r+1\right)-2r^3}\right)\right)=961$ then the value of n is equal to

(A) 31 (B) 30
(C) 60 (D) 61

29. Two tour guides are leading seven tourists. The guides decide to split up. Each tourist must choose one of the guides, but with the condition that each guide must take at least one tourist. Number of different groupings of guides and tourists are

(A) 120 (B) 124
(C) 126 (D) 127

30. If $P(F)=\frac{1}{2}$, $P(G)=\frac{4}{5}$, $P(E\cap F\cap G)=\frac{3}{10}$ and $P(E'\cap F\cap G)=\frac{1}{10}$, the value of $P\left(\frac{F}{G}\right)$ is

(A) $\frac{3}{125}$ (B) $\frac{1}{2}$
(C) $\frac{2}{5}$ (D) $\frac{1}{3}$

Space for Rough work